

# Manisha Kaler

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## EDUCATION

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| <b>University of Massachusetts</b><br><i>Masters of Science in Computer Science</i><br>Coursework: Machine Learning, Performance Engineering, Data Science Fundamentals                                                                                                                                                                        | Jan 2026 - May 2027 (Expected)<br>3.9/4.0 |
| <b>Indian Institute of Technology</b><br><i>Bachelors of Technology in Computer Science and Engineering</i><br>Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Networks, Operating Systems, Computer Organization and Architecture, Algorithms Design and Analysis, Theory of Computation, Computer Graphics | Aug 2017 - May 2021                       |

## WORK EXPERIENCE

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| <b>Arun Data Processing</b><br><i>Software Engineer 2</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Feb 2023 - Dec 2025 |
| <ul style="list-style-type: none"><li>Built and shipped a <b>Go-based PDF generation microservice</b> for the Income Tax Department of India's production tax filing workflow, supporting <b>80M+ taxpayers</b>.</li><li>Engineered <b>schema-driven rendering logic</b> that mapped nested JSON structures to reusable <b>Go templates</b> for tax forms, acknowledgements, and verification PDFs.</li><li>Benchmarked third-party <b>HTML-to-PDF renderers</b> and selected a compatible solution to improve PDF generation reliability for legacy <b>RHEL Linux</b> infrastructure.</li><li>Built a <b>schema-driven code generation pipeline</b> that generated Angular HTML, TypeScript, and service-layer code, reducing new workflow delivery time from <b>4 months to 3 weeks</b>.</li><li>Developed <b>30+ Angular components</b> and shared modules for validation, localization, and <b>WCAG-aligned accessibility</b> across online and offline filing flows.</li><li>Integrated <b>20+ client REST APIs</b> end-to-end and packaged the Angular utility into cross-platform desktop executables using <b>Wails</b>.</li></ul> |                     |

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| <b>VISA</b><br><i>Software Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Jul 2021 - Jan 2023 |
| <ul style="list-style-type: none"><li>Optimized ScalableRun, a <b>global script execution platform</b> managing <b>10,000+ servers</b>, improving platform reliability through <b>real-time validations</b> and <b>asynchronous processing</b>.</li><li>Designed event-driven pub-sub systems using <b>message queues</b> to monitor <b>Ansible Tower</b> automation, surfacing execution metrics in <b>Grafana dashboards</b> to reduce <b>MTTR by 30%</b>.</li><li>Implemented the frontend for <b>Ansible group management</b>, enabling users to organize nested server groups for bulk script execution and improving deployment speed by <b>20%</b>.</li><li>Built an <b>IAM-based access validation layer</b> to enforce user permissions across server groups, ensuring secure script executions across global <b>Cloud PaaS</b> infrastructure.</li><li>Built <b>unit and integration tests</b> for Spring Boot services using <b>JUnit</b>, <b>Mockito</b>, and Spring test utilities, achieving <b>95% code coverage</b>.</li></ul> |                     |

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| <b>VISA</b><br><i>Software Engineer Intern</i>                                                                                                                                                                                                                                                                                                            | May 2020 - Jul 2020 |
| <ul style="list-style-type: none"><li>Developed a full-stack web application using <b>Java</b> and <b>Visa Developer APIs</b> to provide <b>geo-targeted offer recommendations</b> for small merchants.</li><li>Prototyped a functional <b>payroll API</b> solution in Java, digitizing merchant payment workflows for a global case challenge.</li></ul> |                     |

## PROJECT EXPERIENCE

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| <b>Lecture Notes AI Assistant</b>    <i>LLM-based Study Tool</i>                                                                                                                                                                                                                                                                                                                                                         | UMass Amherst |
| <ul style="list-style-type: none"><li>Built an <b>LLM-powered</b> study assistant using <b>Streamlit</b> and <b>Gemini 2.5 Flash-Lite</b> to summarize transcripts, answer grounded questions, and generate study notes.</li><li>Implemented a structured storage layer with <b>SQLAlchemy</b> and <b>SQLite</b> to persist transcripts and support retrieval grounded responses.</li></ul>                                                                                                                 |               |
| <b>Cache-Optimized AVX Matrix Multiplication Kernel</b>    <i>C++, AVX, OpenMP, Linux perf</i>                                                                                                                                                                                                                                                                                                                           | UMass Amherst |
| <ul style="list-style-type: none"><li>Optimized <b>C++ matrix multiplication</b> with loop reordering, cache blocking, <b>AVX vectorization</b>, register-aware kernels, and <b>OpenMP</b>; achieved <b>4.48x</b> single-thread and <b>2.44x</b> 8-thread speedup on large workloads.</li><li>Used <b>Linux perf</b> and assembly inspection to analyze performance, reducing <b>L1 cache miss rate</b> from <b>72.47%</b> to <b>6.56%</b>, then to <b>1.25%</b> with a cache-aware tiled kernel.</li></ul> |               |

## TECHNICAL SKILLS

**Languages:** Go, Java, TypeScript, JavaScript, C/C++, SQL, HTML/CSS, Python  
**Frontend:** Angular, React, Next.js, Bootstrap, Wails, Responsive Design  
**Backend & APIs:** Spring Boot, Gin, RESTful APIs, Microservices  
**AI & LLM:** Prompt Engineering, RAG Pipelines, Gemini API, Grounding  
**Cloud & DevOps:** AWS (EC2, RDS, S3, Lambda), Docker, Jenkins, Ansible Tower, Linux, Git  
**Data & Tools:** PostgreSQL, MySQL, MongoDB, Hazelcast, SQLite, SQLAlchemy, Grafana, perf  
**Testing & Process:** Unit Testing, Integration Testing, Agile/Scrum